

Birla Institute of Technology & Science, Pilani
Work-Integrated Learning Programmes Division
Second Semester 2015-2016
Mid-Semester Test

(EC-2 Regular)

Course No. : SS ZG526
Course Title : DISTRIBUTED COMPUTING
Nature of Exam : Closed Book
Weightage : 35%
Duration : 2 Hours
Date of Exam : 27/02/2016 (AN)

No. of Pages = 2

No. of Questions = 5

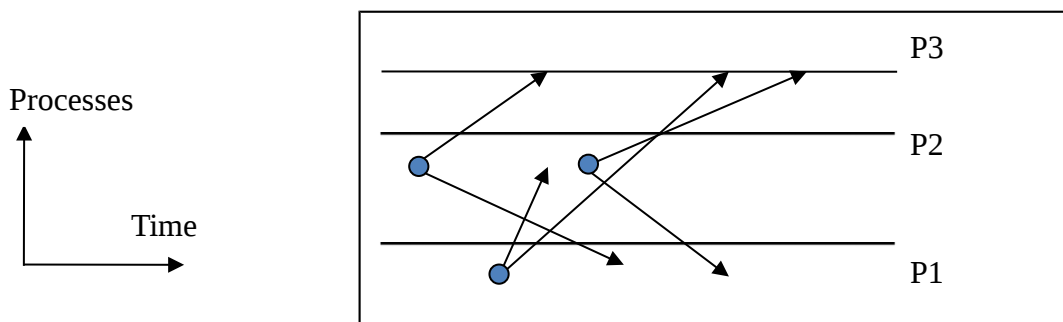
Note:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q.1 (a) Suppose that a client makes only one Remote Procedure Call (RPC) (e.g. to cast a vote). Describe the overheads associated with the use of a binder, i) when the location of the binder is known by the client, and ii) when it is not known. What role a binder does play in RPC? Explain your answer. [2]

Q.1 (b) The state of a communication channel in a consistent global state should be the sequence of messages sent along that channel before the sender's state was recorded, excluding the messages received along that channel before the receiver's state was recorded, i.e, $n \geq m$. Justify this condition with an example. What would be the condition (the given one) for recording a strongly consistent global state? [3]

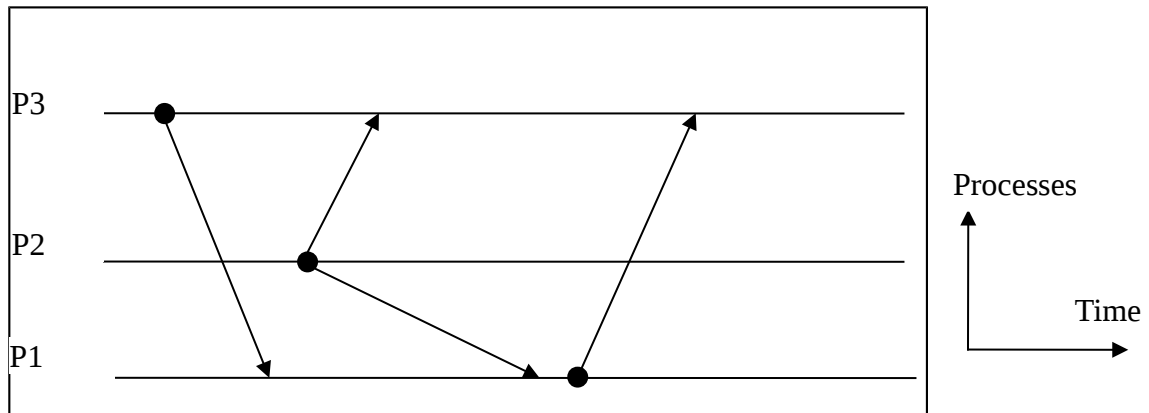
Q.2. Apply Birman-Schiper-Stephenson (BSS) protocol to check if there is any causal order violation in the following time-space diagram depicting three processors and six messages. Show clearly the timestamps and the conditions that may not be true for delivery of some messages. [5]



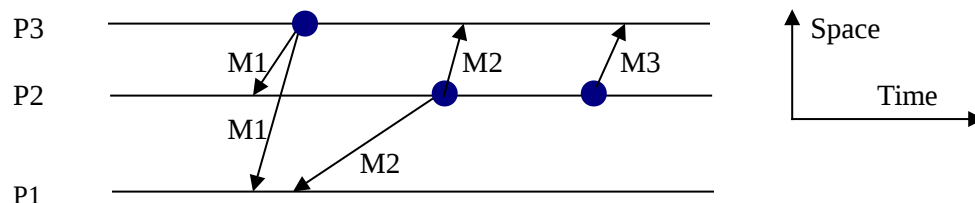
Q.3 (a) What are the main differences between a multiprocessor system and a distributed computing system? Identify any two distributed applications in the scientific and commercial application areas (one each). For each application, list out two design challenges. [3]

Q.3 (b) How is multi-threading on a single processor equivalent to distributed computing in terms of defining what is and what is not simultaneous? How do logical clocks help a distributed or multi-threaded program developer? Explain. [2]

- Q.4 (a) Is it true for Lamport's logical clocks that $C(a) < C(b)$ implies event 'a' happened-before event 'b'? If not, then give the event-trace diagram for a counterexample. [2]
- Q.4 (b) For the following time-space diagram, find out the vector timestamps that you have studied in the course. For the same diagram, also find out the lamport's logical clock values. Which clock is more powerful and why? (Hint: powerful means, it can clearly find out whether two clock values are causally related or not) [8]



- Q.5 (a) Use Schiper-Eggli-Sandoz (SES) causal ordering of messages protocol to find out whether the message sends and receives are causally ordered in the distributed system given below. You are free to do any modifications to the original algorithm if you feel that it is indispensable. [5]



- Q.5 (b) Why is that, solving Mutual Exclusion problem in distributed systems is different from solving the same problem in Multiuser operating systems? Give an example showing the operation of Lamport's Distributed Mutual Exclusion algorithm with number of messages required for Critical Section (CS) invocation as less than $3(N-1)$, where, N is the number of sites in the distributed system. Here, you are allowed to suppress any message (REPLY or RELEASE) but you should insure that the assertion given in the algorithm remains true for allowing a site, entry to the Critical Section. [5]
